
CS246: Database Management Systems Lab

Lab # 11 (1 Questions, 100 Marks)

Lab session: AL1

Held on: 24-Mar-2024 (Mon)

Lab Timings: 14:00 to 17:00 Hours Pages: 3

Submission time: 16:45 Hrs, 24-Mar-2024

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1. This lab theme is centered around embedded SQL in C language.
2. `dbconnect.c` program shows the steps needed to establish connection with database from C language
3. `retrieve-data.c` program shows the steps to fire an SQL and traverse result set
4. Manual pages for stored procedure and stored function are attached.

Question 1: (100 points)

Using C language and MySQL perform the following tasks:

Task 01 (4 marks) Create the database named `week11` using the procedure described in `dbconnect.c`

Task 02 (3 marks) Create the following tables using `mysql_query` function

1. (2 marks) `sailor_name` with the following description:

Column No.	name	data type	Constraints
1	record_number	int	primary key
2	sname	char(20)	

2. (2 marks) `boat_name` with the following description:

Column No.	name	data type	Constraints
1	record_number	int	primary key
2	bname	char(20)	

3. (2 marks) `boat_color` with the following description:

Column No.	name	data type	Constraints
1	record_number	int	primary key
2	bcolor	char(20)	

Task 03 (15 marks) Populate data using `mysql_query` function

1. (5 marks) Populate `sailor_name` table with the data given in the file `sailor_name.csv`.
2. (5 marks) Populate `boat_name` table with the data given in the file `boat_name.csv`
3. (5 marks) Populate `boat_color` table with the data given in the file `boat_name.csv`

Task 04 (3 marks) Create the following tables using `mysql_query` function

1. (1 marks) The **sailors** table has the following description

column #	name	data type	constraint
1	sid	int	primary key
2	sname	char(50)	None
3	rating	int	None
4	age	decimal(3,1)	None

2. (1 marks) The **boats** table has the following description.

column #	name	data type	constraint
1	bid	int	primary key
2	bname	char(50)	None
3	bcolor	char(50)	None

3. (1 marks) The **reserve** table has the following description.

column #	name	data type	constraint
1	sid	int	refers to sailors
2	bid	int	refers to boats
3	day	char(50)	None
sid, bid, day is primary key sid foreign key cascade on update sid foreign key cascade on delete bid foreign key cascade on update bid foreign key cascade on delete			

Task 05 (10 marks) This task to be performed outside C programming language. Create a stored procedure which takes no argument and populates sailor table with 500 sailors by

1. Start the sailor id from 1 and increment by 1 for every new sailor record.
2. Generate first name randomly from the **sailor_name** table
3. Generate age between 18 and 65 (inclusive)
4. Generate rating between 1 and 10 (inclusive)
5. **Hint:** use **RAND()**, **FLOOR()** MySQL functions to obtain the required value

Task 06 (10 marks) This task to be performed outside C programming language. Create a stored procedure which takes no arguments populates **boats** table with information about 50 boats generated as described.

1. Generate a name randomly from the **boat_name** table
2. Generate a color randomly from **boat_color** table
3. Generate **bid** sequentially starting from 1.
4. **Hint:** use **RAND()**, **FLOOR()** MySQL functions to obtain the required value

Task 07 (10 marks) This task to be performed outside the C programming language. Create a stored procedure which takes no argument and populates **reserves** table with 5000 records by

1. Generate a **sid** randomly from the **sailors** table
2. Generate a **bid** randomly from the **boats** table

3. Generate a date between 2025-01-01 and 2025-12-31

4. **Hint:** use `RAND()`, `FLOOR()` MySQL functions to obtain the required value

Task 08 (5 marks) Perform this task outside the C programming language. In-order to generate a date randomly between the specified dates, write a **stored function** which takes no input arguments and has one return value of string data type in the date format `YYYY-MM-DD`. For this consider generating DD part randomly, MM part randomly and YYYY to be 2024. Make sure to check the constraint on DD given MM. For example, if `MM = 02` then DD cannot take values 30 and 31.

Task 09 (15 marks) Call the above stored procedures using `mysql_query` for populating `sailor`, `boat` and `reserve` tables.

Task 10 (20 marks) Create the following tables using `mysql_query` function

1. (10 marks) Retrieve the output of the following query in C language **Find the names of the sailors who have reserved at least two boats.** Firing correct SQL query from C language: 5 marks; Correct output displayed using C language: 5 marks
2. (5 marks) Using `mysql_query`, Create a view having columns `sid`, `age`, `bcolor` for the query **Find the sailors with age over 20 who have not reserved a red boat.**
3. (5 marks) Retrieve data from the view using `mysql_query`.

Task 11 (5 marks) Free the memory allocated for the above queries and close the database connection.